

OS SKILL 13

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    FILE *infile;
    FILE *outfile;

    char text[100];

    printf("=== Input Redirection Test ===\n");

    infile = fopen("input.txt", "r");

    if (infile == NULL) {
        printf("Input file not found\n");
    } else {
        printf("Input file opened successfully\n");

        while (fgets(text, sizeof(text), infile) != NULL) {
            printf("%s", text);
        }

        fclose(infile);
        printf("\nInput stream restored/closed\n");
    }

    printf("\n=== Output Redirection Test ===\n");

    outfile = fopen("output.txt", "w");

    if (outfile == NULL) {
        printf("Unable to create output file\n");
        return 1;
    }
}
```

```

    }

    fprintf(outfile, "Hello from Task 13\n");
    fprintf(outfile, "Output redirection is working\n");

    fclose(outfile);

    printf("Output file created successfully\n");

    printf("\n=== Verify Output Data ===\n");

    outfile = fopen("output.txt", "r");

    if (outfile != NULL) {
        while (fgets(text, sizeof(text), outfile) != NULL) {
            printf("%s", text);
        }

        fclose(outfile);
    }

    printf("\n=== Missing File Test ===\n");

    infile = fopen("missing.txt", "r");

    if (infile == NULL) {
        printf("Missing file handled successfully\n");
    } else {
        fclose(infile);
    }

    printf("\n=== Task 13 Complete ===\n");

    return 0;
}

```

Output:

```
==== Input Redirection Test ====  
Input file opened successfully  
Hello Ubuntu  
This is input redirection  
Task 13 testing  
  
Input stream restored/closed
```

```
==== Output Redirection Test ====  
Output file created successfully  
  
==== Verify Output Data ====  
Hello from Task 13  
Output redirection is working  
  
==== Missing File Test ====  
Missing file handled successfully
```